

Basic Boost Converter — Step-Up DC–DC Conversion

Objective

This note examines the open-loop behavior of a DC–DC boost converter using PSIM simulation.

The focus is on:

- The relationship between duty cycle and average output voltage.
- The influence of inductance, capacitance, and switching frequency on ripple and startup transient response.

The goal is to characterize fundamental boost operation prior to introducing closed-loop control.

System Model

Input voltage: 25 V DC

Load: 5 Ω resistive

Control method: Open-loop PWM

Duty ratios evaluated: $D = 0.3, 0.5, 0.7$

Switching frequency: 5 kHz

All simulations operated in continuous conduction mode (CCM).

Simulations were performed in PSIM to compare output-voltage and inductor-current waveforms under identical hardware conditions.

Basic Principle

Operating Modes

When the switch is ON:

The inductor stores energy from the source, and the diode is reverse-biased, isolating the load.

$$V_L = V_{in}$$

When the switch is OFF:

The inductor's stored energy is released through the diode to the load, adding to the input voltage.

$$V_L = V_{in} - V_{out}$$

The average output voltage is given by:

$$V_{out} = \frac{V_{in}}{1 - D}$$

Example: For $V_{in} = 25V$ and $D = 0.5$, $V_{out} \approx 50V$.

Circuit Parameters

PARAMETER	SYMBOL	VALUE	DESCRIPTION
INPUT VOLTAGE	(V_{in})	25 V	Supply voltage
LOAD RESISTANCE	(R)	5Ω	Represents load
INDUCTOR	(L)	1mH	Controls current ripple
CAPACITOR	(C)	100 μ F	Smooths output voltage
SWITCHING FREQUENCY	(f_s)	5 kHz	Determines ripple and loss
DUTY CYCLE	(D)	0.5	Target ratio for 2 $\times$ boost

These values were selected to maintain continuous conduction while allowing visible ripple and transient effects.

Simulation Observations

Steady-State Behavior

- The average output voltage increased nonlinearly with duty cycle.
- $D = 0.3$ produced approximately 36 V.
- $D = 0.5$ produced approximately 50 V.
- $D = 0.7$ produced approximately 82 V.

The nonlinear gain characteristic follows

$$V_{out} = \frac{V_{in}}{1 - D}$$

which becomes increasingly sensitive as D approaches 1.

Inductor current magnitude increased with duty ratio. Higher D resulted in larger peak and average inductor current.

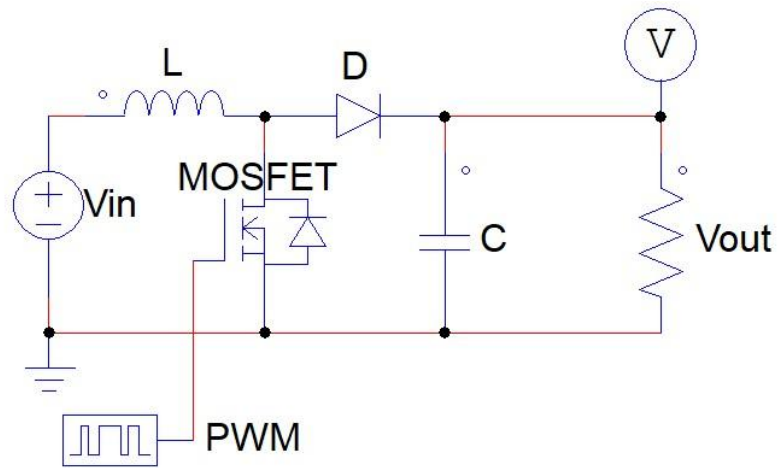


Figure 1

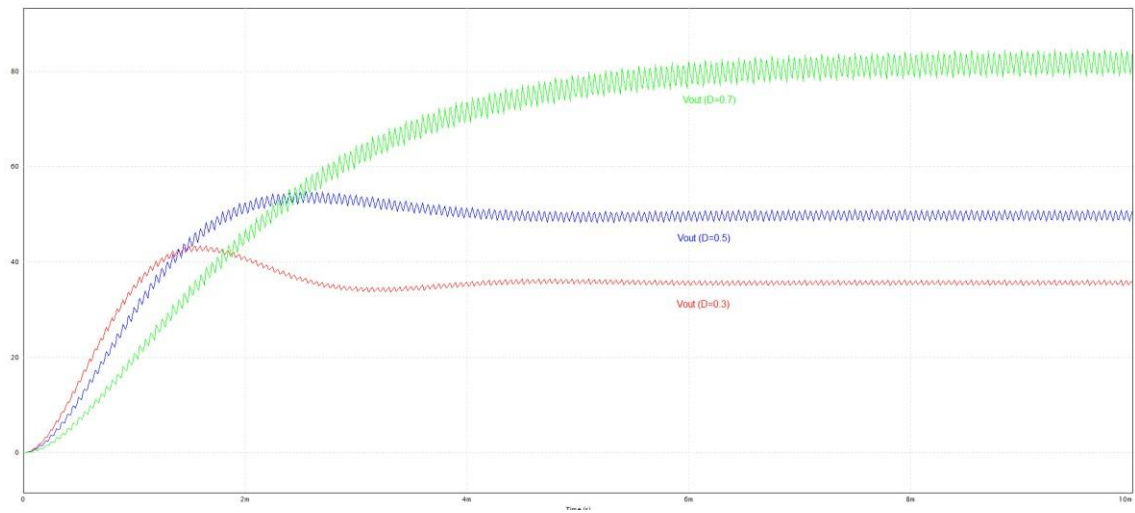


Figure 2

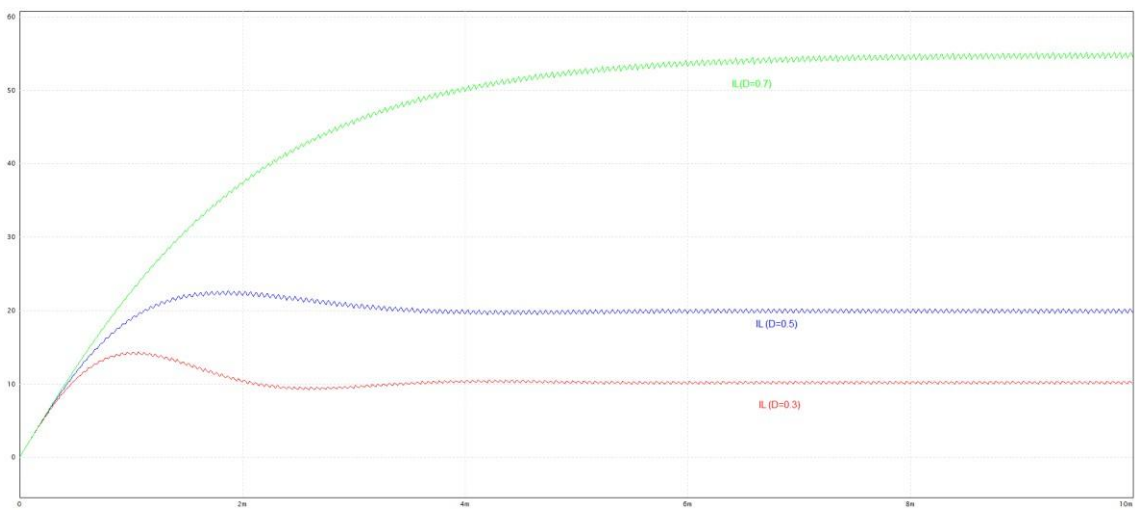


Figure 3

Startup Behavior

- Voltage ripple remained moderate under nominal parameters.
- Rise time increased slightly at higher duty ratios.
- For $D > 0.6$, transient overshoot became more pronounced due to increased stored inductor energy.

This indicates that high-duty operation increases both output gain and energy stress in the magnetic component.



Figure 4

Parameter Sensitivity

Duty Cycle

- Directly determines voltage gain according to :

$$V_{out} = \frac{V_{in}}{1 - D}$$

- High duty ratios significantly increase inductor current and stress.
- Gain sensitivity increases as D approaches unity.

Inductance

- Larger L reduces current ripple.
- Larger L reduces overshoot but slows transient response.

Capacitance

- Larger C reduces output-voltage ripple.
- Excessive capacitance can delay settling if inductor energy release is large.

Switching Frequency

- Higher f_s reduces ripple magnitude.
- Higher f_s improves startup smoothness but may increase switching losses (not evaluated here).

Summary

- In CCM open-loop operation, v_{out} follows

$$V_{out} = \frac{V_{in}}{1 - D}$$

- Voltage gain increases nonlinearly with duty cycle.
- High duty ratios increase both output voltage and inductor current stress.
- Transient overshoot depends on stored magnetic energy and L–C selection.
- Boost converters become increasingly sensitive to duty variation at high D.

This establishes the open-loop boost behavior baseline required before implementing voltage feedback control.